Journal of Anesthesia (2021) 35:854–861
https://doi.org/10.1007/s00540-021-02990-0

ORIGINAL ARTICLE

Intraoperative panic attack in patients undergoing awake craniotomy: a retrospective analysis of risk factors

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Received: 21 April 2021 / Accepted: 14 August 2021 / Published online: 17 August 2021
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Abstract

Purpose Intraoperative anxiety is the most common psychological response of the patient during awake craniotomy. Psychological stress can trigger patient decline, resulting in failed awake craniotomy and significantly poor outcomes. This study aimed to identify the risk factors for panic attack (PA) during awake craniotomies.

Methods With the local ethics committee approval, we conducted a manual chart review of the medical record of patients who underwent consecutive awake craniotomies between November 1999 and October 2016 at Tokyo Women's Medical University. A total of 405 patients were identified and assigned to 2 groups based on the Diagnostic and Statistical Manual of Mental Disorders-V criteria: those that met the PA criteria (Group PA) and those that did not (Group non-PA). Patient characteristics and the incidence of the PA specifier were collected. The features of the two groups were statistically compared, and risk factors for PA occurrence were determined by regression analysis.

Results Sixteen of 405 patients met the diagnostic criteria of PA. Patients' characteristics were not statistically different between the groups. Multivariate logistic regression showed that intraoperative anxiety ($p = 0.0002$) and age younger than 39 years (as opposed to age > = 39 years; $p = 0.0328$) were significantly associated with the occurrence of PA during awake craniotomy.

Conclusions For patients undergoing awake craniotomy, intraoperative anxiety and age younger than 39 years were considered risk factors of PA. As PA often necessitates conversion to general anesthesia, intensive perioperative psychological support and pain management are required to achieve patient satisfaction and the surgical goal of awake craniotomy.

Keywords Anxiety · Awake craniotomy · Complications · Panic attack · Psychological responses

Introduction

Lack of cooperation of the patient due to psychological distress is the second most common intraoperative complication following seizures, during awake craniotomy [1]. While optimal approaches to intraoperative seizures have been identified [2], effective psychological supports for patients are not well established. Psychological responses in awake patients vary widely and anxiety is the most common response [3–5]. In one case of a typical psychological response during awake craniotomy, conversion to general anesthesia (GA) was necessary because the patient became anxious on awakening for cortical mapping and could not be consoled verbally or with pain management [6]. This type of anxiety is caused by frequently observed intraoperative complications such as pain and discomfort [3–5, 7, 8]. Besides, patient anxiety is recognized as the most common warning sign of minor failure of awake craniotomy [4]. A multicenter prospective study found that female patients and

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those of younger age had more frequent complaints of anxiety [3]. However, the occurrence of anxiety is neither center-specific nor a consequence of the anesthesia regimen [7]. Currently, anxiety attack during awake craniotomy is frequently observed but is hardly well managed.

Intraoperative anxiety has psychological impact on a patient who undergoes surgery under awake condition. In the general public, panic disorder is a common anxiety-related syndrome. Panic disorder is characterized by recurrent, unexpected panic attacks (PAs) that are usually accompanied by a variety of somatic and cognitive symptoms. According to the Diagnostic and Statistical Manual of Mental Disorders (DSM)-V, PA is characterized by an abrupt surge of intense fear or intense discomfort that reaches a peak within minutes, and during which time four or more of the following specifiers occur: palpitations, pounding heart or rapid heart rate; sweating; trembling or shaking; feeling of shortness of breath or smothering sensations; chest pain; feeling dizzy, light-headed or faint; feeling of choking; numbness or tingling; chills or hot flashes; nausea or abdominal pains; feeling detached; fear of losing control; fear of dying [9]. However, not all PAs are indicative of panic disorder. The same physical and cognitive symptoms can be provoked when the individual faces the source of fear sensations [10]. Panic attack is often associated with substantial functional deterioration and reduced quality of life. In cases of neurosurgery, tumor recurrence sometimes necessitates multiple operations. Therefore, prevention of psychological trauma formation, especially PA management, is considered to be an important issue for the patients who receive awake craniotomy. Comorbidity of psychological diseases is no longer regarded as an absolute exclusion criterion of awake craniotomy [2, 11]. A cocktail of intravenous sedatives shows benefits to reduce the level of anxiety on some occasions [11, 12]. However, psychiatric reactions evoked by awake craniotomy can occur regardless of the psychiatric history, because fear and pain during operation are not associated with the preoperative psychological condition [13].

Thus, psychological strain during awake craniotomy is still unpredictable and is sometimes uncontrollable. The aim of this retrospective study was to identify the risk factors of PA among patients that underwent awake craniotomies. Psychological determination of PA and identification of potential triggers provide important information on how to tailor the surgical environment for the patient to facilitate successful mapping.

Methods

Ethical approval for this retrospective clinical study was initially provided by the local ethics committee of Tokyo Women's Medical University on November 1, 2016 (No. 4141; Chairperson Prof. Toshimasa Yoshioka), and the study was registered at UMIN Clinical Trial Registry (UMIN000024809, Date of registration: November 11, 2016). Thereafter, the revised protocol, which required a change in the completion date of the study due to delays in analysis, was approved by the local ethics committee of Tokyo Women's Medical University, Tokyo, Japan (No. 4141-R; Chairperson Prof. Toshimasa Yoshioka) on February 8, 2018. Furthermore, as the patients were not subjected to investigational actions and only de-identified retrospective data would be used, the local ethics committee granted a waiver of informed consent. The information to participants about this retrospective study was given via the institution's website before the commencement of the study.

Patients and preparation

Consecutive awake craniotomy cases carried out at Tokyo Women's Medical University between November 1999 and October 2016 were included. To give the patient enough time for preoperative assessment/education and smoking cessation, all cases were planned as elective surgery. Patients were selected based on close anatomical proximity between the lesion and eloquent cortex as indicated by the preoperative magnetic resonance imaging (MRI). These included lesions in the receptive and expressive speech areas in the dominant hemisphere, and motor or sensory cortex in either hemisphere. Following the national guidelines for awake craniotomy, the patients who have already developed moderate or severe symptoms were not selected because functional mapping, as well as neurophysiological monitoring, can be difficult to perform [14]. Preoperative psychological assessment was not a mandatory requirement.

The cornerstone of our strategy is an intensive preoperative assessment and interview with the patient, which were performed by the attending neurosurgeon and anesthesiologist. Before surgery, the purpose of the operation, the procedural process, and any potential complications were explained to the patient, which were supplemented by the scheduled nurses' consultations using a short movie of the entire procedure to provide the virtual experience of the awake craniotomy.

Intraoperative management

No premedication was given. Propofol-based total intravenous anesthesia with intravenous (IV) fentanyl or continuous infusion of remifentanil was employed. Neither nitrous oxide nor neuromuscular blockade use was routinely employed. After the loss of consciousness, the patient's airway was secured by a supraglottic airway (SGA) device. Prior to head fixation, the appropriate head rotation was applied to the supine-positioned patient, and scalp blocks and infiltration

analgesia with long-acting local analgesics (ropivacaine or bupivacaine) were administered. General anesthesia was maintained until the end of the intraoperative MRI (iMRI) scan following the craniotomy as previously described in detail [15]. Once a baseline image was obtained, all administration of anesthetics was ceased. The SGA was removed after the patient regained consciousness.

Surgical intervention was performed using the previously described information-guided technique [16–19]. Our basic surgical plan was to keep the patients awake as long as possible. During cortical stimulation mapping and subsequent resection, sedative-hypnotics were not routinely administered to maintain the required level of consciousness for surgical manipulation, though the decision to provide light sedation was solely made by the attending anesthesiologist. The predetermined analgesic protocol was not defined, thus infiltration analgesia was added for pain management of the scalp, and a small dose of IV fentanyl was given if the patient complained of pain caused by intraoperative positioning. Once the maximum resection of the lesion was confirmed by iMRI, conscious IV sedation was provided for the reminder of surgery per preference of attending anesthesiologist and neurosurgeon. Fentanyl boluses and infiltration analgesia were given if required. Invasive airway management was not applied whenever spontaneous respiration was maintained.

Data acquisition

The retrospective review of consecutive awake craniotomies was done manually to extract data from anesthetic records, nursing charts, and surgical records. During the operations, anesthetic records and nursing charts were completed, whereas surgical records were summarized postoperatively. Patient's vital signs were captured in real time by the automated anesthesia recording system which was introduced in January 2014, although intraoperative events were entered manually by the anesthetist or nursing staff. The sole exclusion criterion was any case where the patient was set up for awake craniotomy before induction but required GA due to unforeseen circumstances. Using the DSM-V classification [9], the included patients were assigned to two groups: those that met the PA criteria (Group PA) and those that did not (Group non-PA).

Data analysis

Patients' characteristics and the incidence of the PA during awake craniotomy were collected. The detailed characteristics of the above 2 groups were compared. Non-parametric data were compared using the chi-square test or Goodness-of-Fit test as appropriate. Parametric data were compared using an unpaired Student's *t* test. Multivariate logistic regression analysis using a stepwise regression with forward selection was used to explore the relationships between PA and patient characteristics. All statistical analyses were performed using JMP® Pro (version 16.0.0, SAS Institute Inc., Cary, NC, USA). Values were expressed as means (SD) unless otherwise specified; $P < 0.05$ was considered to indicate a statistically significant difference.

Results

A total of 407 awake craniotomies were carried out during the study period. One patient manifesting progressive intraoperative hyponatremia, and another patient with the insufficient awake condition due to unforeseen circumstances, were excluded from analysis because GA was maintained throughout surgery based on attending physicians' decision. Of the remaining 405 patients, 16 patients met the diagnostic criteria of PA (Group PA), but 389 patients did not (Group non-PA). The morphometric and demographic characteristics and operative backgrounds of each group are summarized in Table 1. The preoperative characteristics of the patients were not significantly different, other than preoperative physical status. The preoperative condition of the majority of patients was well preserved at American Society of Anesthesiologists Classification of Physical Status (ASA-PS) less than 2, whereas one patient in Group non-PA was classified as ASA-PS 4 due to drug-induced hypersensitivity syndrome. All patients underwent awake craniotomy in the supine position and supplemental oxygen was given through nasal cannula or face mask during the awake phase. The list of cases diagnosed as panic attack was summarized. All patients in Group PA presented four or more of the 16 PA specifiers (Table 2).

Table 1 Characteristics of the patients

	Group PA ($n=16$)	Group non-PA ($n=389$)	*p* value
Age (year)	34 (9)	40 (11)	0.063
Gender (male/female)	10/6	247/142	0.935
BMI (kg/m^2)	23.4 (6.8)	22.8 (3.5)	0.523
ASA-PS (1/2/3/4)	8/6/2/0	184/196/8/1	0.032*
Initial craniotomy	11	271	0.938
Initial awake craniotomy	3	45	0.384
Side of hemisphere (left/right)	12/4	304/85	0.766
Topography (frontal/temporal/parietal/insula)	13/4/0/3	292/71/24/35	0.513

Data are presented as the mean (SD) or as the number of patients or a cumulative number of the patient. *$P < 0.05$

BMI body mass index, *ASA-PS* American Society of Anesthesiologists Classification of Physical Status

Univariate analyses showed no significant association between the incidence of PA and gender ($p=0.935$), body mass index ($p=0.770$), initial craniotomy ($p=0.938$), initial awake craniotomy ($p=0.384$), side of hemisphere ($p=0.766$), childless ($p=0.229$), occupation/schoolwork ($p=0.438$), intraoperative low back pain ($p=0.504$), intraoperative seizure ($p=0.660$) or intraoperative nausea/vomiting ($p=0.509$). However, age ($p=0.0108$), ASA-PS ($p=0.0140$), marital status ($p=0.0405$), intraoperative headache ($p=0.0344$), or intraoperative anxiety disorder ($p<0.0001$) were significantly associated with intraoperative PA (Table 3). Similarly, multivariate logistic regression analyses using a stepwise regression showed that intraoperative anxiety disorder ($p=0.0002$) and age less than 39 years (median = 38 years; $p=0.0328$) were significantly associated with a higher incidence of PA during awake craniotomy with 0.1800 of R^2 and 118.61 of Akaike's Information Criterion (AIC) (Table 4). Seven patients of Group PA received intravenous sedatives based the attending anesthesiologists' decision, of which 3 patients (0.7%) required GA conversion due to emotional intolerance that occurred subsequent to PA.

Discussion

This study identified intraoperative anxiety and age less than 39 as risk factors for PA during awake craniotomy. These findings help to provide further clarity regarding risk factors that need to be considered to prevent PA during awake craniotomy and thereby reduce the likelihood of conversion to GA.

Intraoperative psychological responses during surgical intervention need to be minimized as these cause a significant decline in awake patients. Conversion to GA due to unexpected circumstances correlates with poor outcomes. Nossek et al. reported that failed awake craniotomy leads to significantly lower gross-total resection rate and longer hospitalization [20]. In our case series, 16 of 405 patients (4.0%) met the diagnostic criteria of PA, of which three patients (0.7%) required GA because emotional intolerance subsequently occurred to PA. As awake craniotomy has become a well-tolerated procedure and overall patient satisfaction is high, failure of awake craniotomy, defined as the inability to complete intraoperative functional mapping or the need to convert to GA, is becoming a rare complication, occurring in fewer than 2% of cases irrespective of anesthesia technique [21]. Our GA conversion rate of 0.7% is considered to be reasonable because factors other than PA are also common reasons for urgent intervention. It is notable that the surgery-related pain triggered PA development in all three GA-converted patients in our study. One patient could not endure persistent pain at the pin holding sites. The extended surgical intervention to the middle fossa floor induced intractable pain, which is similar to trigeminal neuralgia, observed in the other two patients. Studies show that there are significant number of awake patients who experience intraoperative emotional distress due to pain [7]. Theoretically, the principles of pain management for awake craniotomy are effective regional analgesia and opioid use. Scalp blocks along with infiltration analgesia for incision line and pinning site using long-acting local anesthetics can provide reliable pain relief for the majority of patients. Opioids remain the main reserve analgesic agent but could cause intraoperative nausea/vomiting or somnolence [22]. However, intraoperative pain is sometimes uncontrollable. As our two GA-converted cases indicate, surgical manipulation-induced pain is relatively common but hard to control. In particular, the dura mater of the skull base, falx cerebri, and the leptomeninges of the lateral fissure and neighboring sulci are recognized as the pain-sensitive intracranial structures [23]. Pain of these areas is problematic because supplemental local analgesics are not effective and opioids sometimes affect cognitive function. Therefore, the preventative measure of this phenomenon is to not make contact with the pain-sensitive intracranial structures while the patient is awake. The whole roadmap of the surgery should be shared before operation by not only neurosurgeons, but also other clinical staff, including anesthesiologists and nurses. If manipulations of the skull-base structures or perivascular traction are considered to be frequently required, the intentional delay of transition from the pre-awake to the awake phase is an alternative option to reduce the levels of pain and discomfort, hence the risks of PA and GA conversion.

Panic disorder is a common anxiety disorder with lifetime prevalence among Americans of 7.0% in women and 3.3% in men [24]. Despite the prevalence variability across countries, a higher frequency of panic disorder in female individuals has been shown consistently [10]. In the present study, the female gender was not determined as a PA risk factor; but two of three GA-converted patients in the Group PA were female. It cannot be denied that there is a possible bias in our cohort as they were highly motivated and were willing to receive their craniotomy awake. They had sufficient time to comprehend the reality of the operation through the preoperative visits using a short movie of the entire procedure. However, the fact that female patients more frequently complain of intraoperative anxiety [3], suggests that female gender is potentially more prone to PA than male gender, even though awake craniotomy is an entirely different environment from the ordinary. Another concern about intraoperative PA is the potential risk of posttraumatic stress disorder (PTSD). Compared to the patients suffering from cancer of other body sites, the patients harboring brain tumors suffer from a higher rate of mental distress and are more prone to experiencing anxiety and/or depression [25, 26].

Table 2 List of cases diagnosed as panic attack

Patient no.	Palpitations, pounding heart, or accelerated heart rate	Sweating	Trembling or shaking	Sensations of shortness of breath or smothering	Feelings of choking	Chest pain or discomfort	Nausea or abdominal distress	Feeling dizzy, unsteady, lightheaded, or faint	Chills or heat sensations	Paresthesias (numbness or tingling sensations)	Derealization (feelings of unreality) or depersonalization (being detached from oneself)	Fear of losing control or "going crazy"	Fear of dying
1	Yes	No	Yes	Yes	Yes	No	No	No	No	No	No	Yes	No
2	Yes	No	No	No	Yes	No	Yes	No	No	No	No	Yes	No
3	Yes	No	Yes	Yes	No	No	Yes	No	No	No	No	No	No
4	Yes	No	No	Yes	Yes	No	No	No	No	No	No	Yes	No
5	Yes	No	Yes	Yes	No	No	No	No	No	No	No	Yes	No
6	Yes	Yes	No	Yes	No	No	No	No	Yes	No	No	Yes	No
7	Yes	No	No	Yes	Yes	No	No	No	No	No	No	Yes	No
8	Yes	No	No	Yes	Yes	No	No	No	No	No	No	Yes	No
9	Yes	No	No	Yes	No	No	No	No	No	No	Yes	Yes	No
10	Yes	No	Yes	Yes	No	No	Yes	No	No	No	Yes	Yes	No
11	Yes	No	Yes	Yes	No	No	No	No	No	No	No	Yes	No
12	Yes	No	No	Yes	No	No	Yes	No	No	No	No	Yes	No
13	Yes	No	Yes	Yes	No	No	No	No	No	No	No	Yes	No
14	No	No	No	Yes	Yes	No	Yes	No	No	No	No	Yes	No
15	Yes	No	No	Yes	No	No	Yes	No	No	No	No	Yes	No
16	Yes	No	Yes	Yes	Yes	No	No	No	No	No	No	Yes	No

Table 3 Univariate analysis of panic attack during awake craniotomy

	Group PA ($n=16$)	Group non-PA ($n=389$)	OR (95% CI)	p value
Patient characteristics				
Age (year)				
< 39[a]	12	194	0.177	0.0108*
≧39	4	195	(0.040–0.787)	
ASA-PS				
1 or 2[a]	14	380	6.032	0.0140*
3 or 4	2	9	(1.191–30.551)	
Marital status				
Married	7	252	0.361	
Single[a]	9	117	(0.131–0.993)	0.0405*
Gender				
Male	10	247	0.958	0.935
Female[a]	6	142	(0.341–2.692)	
BMI (kg/m^2)				
< 22[a]	8	180	0.861	0.770
≧22	8	209	(0.317–2.341)	
Initial craniotomy				
Yes[a]	11	271	1.044	0.938
No	5	118	(0.355–3.071)	
Initial awake craniotomy				
Yes	3	45	1.764	0.384
No[a]	13	344	(0.484–6.430)	
Patient characteristics				
Side of hemisphere				
Left	12	304	0.839	0.766
Right[a]	4	85	(0.264–2.667)	
Child				
Have	6	195	0.535	0.229
Do not have[a]	10	174	(0.191–1.503)	
Occupation/schoolwork				
Have	14	292	1.798	0.438
Do not have[a]	2	75	(0.400–8.083)	
Intraoperative complications				
Headache				
Yes	11	155	3.037	0.0344*
No[a]	5	214	(1.034–8.919)	
Anxiety disorder				
Yes	10	40	8.725	< 0.0001*
No[a]	6	349	(3.105–24.517)	
Low back pain				
Yes	1	13	2.024	0.504
No[a]	13	342	(0.246–16.658)	
Seizure				
Yes	2	69	0.713	0.660
No[a]	13	320	(0.157–3.234)	
Intraoperative complications				
Nausea and vomiting				
Yes	6	124	1.425	0.509
No[a]	9	265	(0.496–4.091)	

Data are presented as the number of patients. *$P < 0.05$

[a]Reference category

OR odds ratio, *CI* confidence interval, *BMI* body mass index, *ASA-PS* American Society of Anesthesiologists Classification of Physical Status

Table 4 Multivariate logistic regression model predicting panic attack during awake craniotomy

	Coefficient	OR	(95% CI)	*p* value
Age (year)				
≧39 vs < 39[a]	− 0.827	0.191	0.042–0.874	0.0328*
ASA-PS				
1–2[a] vs 3–4	0.925	6.357	0.922–43.802	0.0604
Anxiety disorder				
Yes vs no[a]	1.025	7.772	2.666–22.658	0.0002*

Statistical analysis was performed with multivariate logistic regression analysis using a stepwise regression with forward selection. *$P < 0.05$

OR odds ratio, *CI* confidence interval

[a]Reference category

Moreover, awake craniotomy becomes an additional traumatic experience with possible negative consequences on mental health. There are patients who had significant fear or even PTSD-like symptoms after awake craniotomy [7, 27, 28]. In a study by Goebel and colleagues, one of 25 patients complained about reexperiencing noises during craniotomy as intrusions after the procedure [29]. Milian et al. investigated the presence or absence of PTSD symptoms in patients who underwent awake craniotomy, and found that 12.5% of their patients had nightmares after surgery. The study also reported that younger age at surgery and female gender were risk factors for PTSD symptoms of increased arousal during sleep [7]. Our cohort showed that 14 patients (3.5%), of which 2 patients in the Group PA and 12 patients in the Group non-PA, required postoperative psychiatry liaison due to newly manifested psychological symptoms. Although the detailed psychological sequelae were not examined in this retrospective study, one of our PA risk factors of younger age is in concordance with the PTSD risk factor identified by Milian et al. [7]. In comparison to the older generation, younger patients are more likely to have a greater number of subsequent craniotomies due to tumor recurrence. Therefore, it is considered that young patients who undergo awake craniotomy will be the main target of intensive perioperative mental care, not only for reducing intraoperative PA, but also for avoiding PTSD development. Actually, the prospective observation conducted by the European Low Grade Glioma Network presented that three of 15 patients who underwent awake craniotomy stated that they would prefer GA management instead of such a procedure again [3]. Goel et al. also reported that 88% of the patients would participate in craniotomy in awake state again if necessary, whereas 12% were not sure [29]. With the remarkable advancements of surgery and adjunct therapies, the demand for multiple operations for tumor recurrence has been increasing. Making psychological trauma on the patient during the awake craniotomy should be minimized because it will be a significant disadvantage to the patient if the subsequent awake craniotomy cannot be accepted due to mental strain.

Our study has limitations inherent in a single-center retrospective design, including selection and follow-up biases. Although the database was constructed on a prospective basis, the chart review process was retrospectively performed, and thus not all related factors could be collected. For instance, intraoperative anxiety was identified by manual chart review of the medical record by extracting the annotation of "anxiety" made by the clinical staff's impressions. The incidence of anxiety in the current study is possibly underestimated if the clinical staff did not give complete attention to the patients. The patient's postoperative psychological sequelae were also not systemically followed. The clinical impact of PA on surgical outcomes and the appropriate regimen of PA treatment are considered to be the future subjects. As the study is based on the single-center experience, the learning curve toward patient care may have influenced surgical performance. However, the standard of patient care and surgery is considered to be universal. To the extent that similarities can be found with other institutions, the associations derived from the current study will inform the daily decision-making process during awake craniotomy.

In conclusion, anxiety disorder, a major psychological response that occurs in awake patients, is also a risk factor for intraoperative PA. Although PA is a rare complication of awake craniotomy, it potentially becomes a reason for GA conversion. Therefore, especially for patients of younger age at surgery, intensive perioperative psychological support and pain management are required to achieve patient satisfaction and the surgical goal of awake craniotomy.

Acknowledgements Dr. Takashi Maruyama helped to design the study and to write the manuscript. Dr. Ryu Komatsu helped to interpret the data and provided valuable expertise in writing the manuscript. Dr. Makoto Ozaki helped to design the study and helped to write the manuscript. The authors would like to acknowledge Dr. Yoko Uchiide (Certified psychiatrist) for giving a wide range of views in the psychiatric field. The authors are also indebted to Dr. Satoru Shimizu (School of Arts and Sciences, Tokyo Woman's Christian University) for his expert advice about statistical analysis. The authors would also like to acknowledge Dr. Yoshihiro Muragaki (Department of Neurosurgery, Faculty of Advanced Techno-Surgery, Institute of Advanced Biomedical Engineering and Science, Tokyo Women's Medical University), Dr. Takakazu Kawamata (Department of Neurosurgery, Tokyo Women's Medical University), Dr. Masayuki Nitta (Department of Neurosurgery, Tokyo Women's Medical University), Dr. Taiichi Saito (Department of Neurosurgery, Faculty of Advanced Techno-Surgery, Institute of Advanced Biomedical Engineering and Science, Tokyo Women's Medical University), and Dr. Manabu Tamura (Department of Neurosurgery, Faculty of Advanced Techno-Surgery, Institute of Advanced Biomedical Engineering and Science, Tokyo Women's Medical University) for their valuable cooperation.

Declarations

Conflict of interest The authors report no financial interests or potential conflicts of interest. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors. Preliminary data for this study, which the local ethics committee approved of Tokyo Women's Medical University on November 1, 2016 (No. 4141; Chairperson Prof. Toshimasa Yoshioka), was previously presented as a poster presentation at the Euroanaesthesia 2017 Annual Meeting, 3–5 June 2017, Geneva, Switzerland.

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